

CS 30700: Product Backlog

BoilerSpace

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Problem Statement

Purdue students struggle to find open study spaces in busy buildings like Lawson, WALC, or Krannert because there is no way to check room availability without physically walking to the location, sinking 30 to 60 minutes walking between buildings during peak hours. Additionally, students often miss out on helpful academic collaboration because there is no central hub to schedule future study sessions or access notes from previous group meetings. Our project, BoilerSpace, is different because it combines a live crowdsourced room map with a permanent digital library and scheduling tool. Unlike the Purdue room scheduling system that is generic and does not tailor a studying experience, or platforms like GroupMe that focus on real-time messaging that specific course collaboration, our platform allows users to search through past session notes using an AI chat, congregated by a multitude of students from the class, and a view "live vibes" of campus buildings to decide where to work with likeminded perspective students.

Background Information

Problem, domain, and targeted users (Audience):

Purdue classes seem to get harder by the semester, so students have to find better ways to study for these classes. What better way to do this than to collectively study as a class? In the age of AI, students will love our AI notes condensation feature as well as being able to meet new students studying for the same class, trying to achieve the same goals. Our messaging and map UI will be directly targeted at college students preparing for life changing exams.

Similar Platforms:

Some similar platforms to BoilerSpace are GroupMe and Discord due to the user messaging features where users can create conversations by a topic/class (threads/group chats). Notion is also a similar platform since it offers users the ability to collaborate with others in a shared workspace and upload materials.

Limitations:

BoilerSpace stands out for various reasons. For one, official Purdue room scheduling platforms and physical solutions only show formal bookings, not real-time usage. A room might be "reserved" or "booked" but turn out to be empty the whole time or full with squatters. Students currently resort to walking between different buildings to check for rooms, tables, and chairs, wasting 30-60 minutes during peak hours. We tackle this problem with a crowdsourced "live vibe" check-in system. Users can report real-time capacity and noise levels, allowing students to check room availability remotely without needing to walk around campus.

Additionally, BoilerSpace stands out among platforms like GroupMe, Discord, and Snapchat for messaging because they are not designed for making student resources, past exam solutions, and meeting notes quickly accessible, as they can get buried in a sea of messages. They also lack location-based context for meetups. Our

platform utilizes threaded study sessions and a digital library. Our AI integration indexes shared notes, allowing students to search for concepts instead of needing to scroll through chat logs.

In terms of a map, Google Maps is great for general "popular times" data, but it lacks granular, room-level data. It cannot tell if a specific floor is full or if a room with a whiteboard is available, for example. We provide an interactive campus map that is detailed at the room level. It provides details about amenities like whiteboards, outlets, silence enforcement, and more.

Finally, platforms like Notion and Google Drive are great for personal or small groups, as they require specific invites to access. However, this prevents students from discovering open study groups or widely shared class resources. BoilerSpace is building an open platform where study sessions are discoverable by class or major, breaking down the silos that prevent students from finding study partners or rich study materials.

Functional Requirements:

Login/Profile:

1. As a user, I would like to register for a BoilerSpace account so that I can save my preferences and access the full suite of features.
2. As a user, I would like to be able to login and manage my BoilerSpace account so that I can keep my personal data secure and up to date.
3. As a user, I would like my password to be reset if I forget it so that I can regain access to my account without creating a new one.
4. As a user, I would like to be able to verify my Purdue email by getting an OTP so that the platform remains exclusive to the campus community and prevents spam.
5. As a user, I would like to be able to edit my profile (display name, major, year) so that others recognize me in sessions.
6. As a user, I want to restrict/block another user, so that I don't see their sessions or messages.
7. As a user, I want to set my public/private profile visibility so that I control what others see.
8. As a user I want to set my classes for this semester, so that others know what I am studying
9. As a user, I want to be able to friend others based on what classes they are taking so that I can build a network of study partners for my specific coursework.
10. As a user, I want to update my availability or preferred study times, so that others know when I'm typically open to studying.
11. As a user, I want to manage my notification preferences from my profile, so that I can control how and when I'm contacted.
12. As a user, I want the option to upload a profile picture so others can recognize me more easily.
13. As a user, I want to be able to delete my account so that I can remove my personal data if I leave the university or stop using the app, and we will also have 2FA.

Insights

14. As a user, I want to see my total study time for this week, so that I can track my productivity.
15. As a user, I want to receive achievements for hitting study goals so that I stay motivated to maintain a consistent study routine.
16. As a user, I want AI recommendations based off my study habits so that I can improve efficiency
17. As a student, I would like to receive notifications for when a room is under a certain capacity so that I can travel to the location knowing I will successfully find a seat.
18. As a student, I want to see a clear explanation when location access is requested, so that I understand why it is needed before granting permission.
19. As a student, I want to receive reminders for upcoming study sessions I joined, so that I don't forget to attend.

20. As a student, I want to receive notifications when new content is uploaded to courses, sessions, or events that I follow so that I stay up to date without manually checking the platform.
21. As a student, I want to temporarily mute all notifications, so that I can focus without interruptions.
22. As an admin, I want to deactivate abusive accounts so that the community stays safe, if someone gets reported too many times, we will remove them from our platform.
23. As an admin, I want to verify official campus organizations, so that students can trust event legitimacy.
24. As an admin, I want to remove inappropriate content (notes, comments, or events), so that the platform remains respectful.
25. As a student, I want to confirm or dispute a previous user vibe report so that the live data remains accurate and up to date
26. As a student, I want to see a timestamp of last update for a room status so I can judge how reliable the information is before I start walking
27. As a student, I want to filter the map for quiet zones specifically so I can find a place for deep focus rather than group work.
28. As a student, I want to export an AI generated study guide to PDF or markdown so I can study offline or print it out
29. As a student, I want to see a simple green or red dot next to a room name so I can instantly know if it is likely open or full
30. As a student, I want to click a single button to check in to a room so I can contribute to the live data without filling out a long form
31. As a student, I want to see a basic list of buildings sorted alphabetically so I can find my favorite study spot quickly
32. As a student, I want to see how many people are currently checked into a specific building so I can gauge the overall crowd level
33. As a student, I want to use a search bar to find a specific building by its name or campus abbreviation
34. As a student, I want my dashboard and room pages to display recent buildings I've visited, key room amenities (such as whiteboards and outlets), and the last updated time of room status so that I can quickly choose a study space that fits my needs and trust the availability information.
35. As a student, I want to filter my course list by the current semester so I can focus only on the classes I am taking right now
36. As a student, I want to see a small icon next to a building name that indicates if it has a cafe or vending machines nearby so I can choose a study spot based on snack availability
37. As an admin, I want automatic spam detection so less moderation work is needed
38. As an admin, I want to be able to broadcast announcements so users can receive platform updates
39. As an admin, I want to see platform analytics to see which features are most popular, least-used, and make data-driven decisions to improve the platform
40. As a student looking for a seat, I want to view an interactive campus map so that I can quickly see buildings and rooms.
41. As a student I want directions to different study locations so that I can find the fastest route to a building I haven't visited before.
42. As a student, I want to see a list of rooms sorted by distance from my current location so that I can find the nearest available spot.
43. As a student, I want to bookmark my favorite rooms so that I can quickly check them in the future.
44. As a student I would like to find people from my class to study with so that we can help each other understand course material.
45. As a student, I want to search for certain classes that people are studying for so that I can join a group that is working on the same assignment/project as me.
46. As a student, I want to search for certain clubs and if they are hosting any study get-togethers so that I can meet people with shared interests.

47. As a student, I want to comment on a session's notes so that I can ask clarifying questions or provide corrections to shared content.
48. As a student, I want to know certain exam dates for certain classes so that I can prioritize my study sessions effectively.
49. As a student, I want the option of dark mode so that I can study more comfortably at night.
50. As a student, I want to adjust the text size, so that all the content is easier to read.
51. As a student, I want to view and upload notes from study sessions for my course so that I can both review material I may have missed and contribute to the shared knowledge base.
52. As a student, I want to upvote/downvote certain notes so that the most helpful and accurate resources rise to the top.
53. As a student, I want to rate and post reviews on study areas so others can find the best spots
54. As a student, I want to filter rooms based off amenities (whiteboards, outlets) so that I can find a workspace that fits my equipment needs.

AI Assistant

55. As a student, I want to ask AI assistant questions about the aggregated study notes for my course so that I can find specific explanations without reading every document.
56. As a student, I want the AI assistant to show citations that link back to the exact note or page it used, so that answers are traceable and verifiable.
57. As a student, I want the AI assistant to generate concise study guides from my notes and highlight key topics, so I can review efficiently and know what to focus on.
58. As a student, I want the AI to suggest practice questions without giving full solutions, so that I can test my understanding.
59. As a student, I want to bookmark helpful AI responses, so that I can easily return to them later.
60. As a student, I want the AI to help create a study plan and schedule so I can organize my time efficiently
61. As a student, I want AI to suggest external resources like textbooks or videos to improve my understanding of certain topics

Clubs / Organizations

62. As a club organizer, I want to post events with time and location so that students can discover them easily.
63. As a club organizer, I want to tag events by category (callout, workshop, social, fundraiser) so that students can filter them.
64. As a club organizer, I want to create an event with a title, description, date, time, and campus location, so that students clearly understand what the event is.
65. As a club organizer, I want to manage who can create events under my club so that only authorized members can post.
66. As a club organizer, I want to attach external links (sign-up forms, Discord, social media) so that students can engage beyond the event.
67. As a club organizer, I want to see how many students plan to attend so that I can plan resources.
68. As a club organizer, I want to post announcements or updates related to events, so that attendees stay informed.
69. As a club member I want to RSVP to events so that the organizers know to expect me.
70. As a club organizer, I want to restrict even visibility to certain audiences (member-only or public), so that events are targeted.
71. As a club organizer, I want to create and manage a club profile, so that students can learn about the organization.

- 72. As a club organizer, I want to assign different roles to club members (e.g., admin, officer) so that responsibilities are clear.
- 73. As a club organizer, I want to approve or remove club members, so that organizations stay moderated.
- 74. As a club organizer, I want to schedule recurring events so members can plan attendance in advance
- 75. As a club member, I want to see files or slides from past events to stay up to date
- 76. As a club organizer, I want to see past attendance statistics so I can evaluate engagement

Communication

- 77. As a student, I want threaded discussions for session pages, so that conversations are organized. (if time allows)
- 78. As a student, I want to reply to specific messages within a session, so that side conversations stay contained within the main discussion thread.
- 79. As a student, I want to receive notifications when someone replies to a thread I participated in, so that I can stay engaged without monitoring the entire chat.
- 80. As a student, I want to mute specific threads within a session, so that I can focus on conversations that matter to me.
- 81. As a student, I want to see a timestamp on messages in session discussions, so that conversations have context.
- 82. As a student, I want to search within a session's discussion threads, so that I can find specific information.
- 83. As a student, I want to react to messages with simple reactions, so that I can acknowledge messages without replying.
- 84. As a student, I want to be able to delete messages if I accidentally send something and have a message appear when I do so that others know I deleted a message

Non-Functional Requirements

Architecture and Performance:

BoilerSpace will use a client server architecture based on the MERN stack (MongoDB, Express, Node.js) to support maintainability and future growth. The frontend will be a responsive web application built with React and Tailwind CSS, while the backend will provide a REST API built with Node.js and Express to manager user accounts, room availability, study sessions, file uploads, and AI-assisted queries. This separation allows system components to be developed and updated independently.

The system will be designed to provide responsive performance under typical usage. Common actions such as viewing room availability, browsing study sessions, and loading the campus map should be completed within 1 second. More resource intensive actions including operations with the AI Knowledge Chatbot should be completed within 3 seconds. The application is expected to support approximately 300 concurrent users without noticeable performance degradation.

Authentication and authorization will be implemented using Passport.js within our Express backend. User passwords will be salted and hashed using bcrypt (with a work factor of 10) before storage in MongoDB. To secure the REST API, we utilize the passport-jwt strategy, requiring a valid, signed JSON Web Token or JWT in the authorization header for all protected endpoints. This stateless authentication method prevents Cross-Site Request Forgery and ensures data scraping is blocked.

Crowdsourced Data Integrity (Security):

BoilerSpace relies on crowdsourced check-ins to provide live room availability and study environment indicators. Room data should be based on combined input from multiple users, while still allowing recent individual check-ins to influence results when limited data is available. The system should account for users leaving rooms by gradually reducing the impact of older check-ins over time, ensuring that availability indicators stay aligned with real campus conditions.

Usability:

The BoilerSpace interface emphasizes clear organization to help users efficiently discover study spaces, sessions, and campus events. The interactive campus map enables users to explore locations, view ability, and access relevant information with minimal effort. Study sessions, courses, and club events are structured with filtering and search options to make content easy to find and sort. The AI chatbot is designed to be easily accessible within the application, allowing users to ask questions and review course related content without disrupting their workflow. Users should also be easily able to view past notes and materials shared by other students or generated through the AI assistant. Profile features such as class selection and visibility controls are streamlined so users can manage how they are represented to others. The application will provide a consistent and usable experience across devices, screen sizes, and modern web browsers.

Hosting/Deployment:

To maintain continuous delivery and scalability, the application will use a split deployment strategy. The frontend will be hosted on Vercel, which provides automatic deployments from our GitHub repo and a global CDN for fast load times. The backend will be deployed on Render. This way we can implement our crowdsourcing and messaging features, which require a long-running server process.